	AIR UNIVERSITY
	DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
	EXPERIMENT NO 2

Lab Title Implementation of Loops & Selection Control Structure

Student Name: Ali Abbas Khan **Reg. No:** 230661

Objective:

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____



Department of Electrical and Computer Engineering

Signal and Systems Lab

Lab Report: 2

Title: Introduction to

Loops and Selection

Control Structures

Name: Ali Abbas Khan

Roll Number: 230661

Class: BCE-5B

Submitted To: Engr. Maryam
Date: September 28, 2025

Lab Tasks:

1. Use a for loop to generate a table of 5
Note : the number should be input by the user

Code:

Editor - C:\Users\Admin\OneDrive\Desktop\SNS Lab\Matlab Files\L2.m

L2.m

```
clc;
A=input('Enter a Number: '); %Task 1
j=1;
while j<=10
    p=A*j;
    fprintf("%d x %d = %d\n",A,j,p);
    j=j+1;
end
```

Output:

Editor - C:\Users\Admin\OneDrive\Desktop\SNS Lab\Matlab Files\L2.m

Command Window

New to MATLAB? See resources for Getting Started.

```
Enter a Number: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

2. Takes a positive integer as input.
The mod(num, 2) function checks the remainder when the number is divided by 2.
• If the remainder is 0, the number is even.
• If the remainder is not 0, the number is odd.

Code:

```
B=input('Enter a Number: ');  %Task 2
d=mod(B,2);
if d==0
    fprintf("%d is Even\n",B);
else
    fprintf("%d is Odd\n",B);
end
```

Output:

```
Enter a Number: 6
6 is Even
```

3. Declare a vector A consisting of 9 elements. Use for loop to find the product of all the elements of the vector and store the result in a variable ‘z’. Subtract 12 from z and store the final result in “c”.

Code:

```
18 -    A=input('Enter a vector of 9 elements: ');
19 -    z=1;
20 -    for i=1:9
21 -        z=z*A(i);
22 -    end
23 -    fprintf("Z=%d\n",z);
24 -    c=z-12;
25 -    fprintf("c=%d\n",c);
```

Output:

```
Enter a vector of 9 elements: [1 2 3 4 5 6 7 8 9]
Z=362880
c=362868
```

4. Write A equation in Matlab

cos(2x)+sin(3y)

log2(x)+e^{3x}

Where x <5 and y=3

Code:

```
27 -    x=0:0.5:4.9;
28 -    y=3;
29 -    g=cos(2*x)+sin(3*x);
30 -    fprintf("g=%d \n",g);
31 -    h=log2(x)+exp(3*x);
32 -    fprintf("h=%d \n",h);
```

Output:

```
g=1
g=1.537797e+00
g=-2.750268e-01
g=-1.967523e+00
g=-9.330591e-01
g=1.221662e+00
g=1.372289e+00
g=-1.257935e-01
g=-6.820730e-01
g=-1.073458e-01
h=-Inf
h=3.481689e+00
h=2.008554e+01
h=9.060209e+01
h=4.044288e+02
h=1.809364e+03
h=8.104669e+03
h=3.631731e+04
h=1.627568e+05
h=7.294185e+05
```

Conclusion:

This lab provided practical exposure to core programming structures in a computational environment like MATLAB or Octave. We successfully implemented a multiplication table using a while loop, demonstrated conditional logic with an if-else statement to determine if a number is even or odd, and utilized a for loop to calculate the product of elements within a user-defined vector. These tasks solidified understanding of user input, formatted output, iteration, conditionals, and the efficient use of built-in mathematical functions for array manipulation.